PLANT 2 Standard Operating Procedure	Size Changeover, Line 2 (2 L and 500 ml)		
	Document No.: **SOP-104**	Revision: **B**	Effective: **2021-03-15**
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1.0 PURPOSE

Change Line 2 between the 2 L and the 500 ml fabric softener bottle.

2.0 SCOPE

Line 2 only. Products FS-2000 and FS-0500.

3.0 RESPONSIBILITIES

Role	Responsibility
Shift Lead	Releases the line for changeover and signs the record.
Operator	Carries out the changeover.
Maintenance Technician	Fits the filler change parts.
QA Technician	Gives first-off approval.

4.0 SAFETY AND MATERIALS

Personal protective equipment: Safety glasses; Safety shoes; Cut-resistant gloves

Materials: Bottles; Caps (38 mm); Labels; Shipper cases

Tools: Change parts kit (stored in the maintenance crib); Torque tester; Hex key set; Feeler gauge

5.0 RELATED DOCUMENTS AND RECORDS

Documents: SP-402 Lockout/Tagout; WI-210 Filler Setup and Nozzle Check; WI-212 Labeler Setup and Label Roll Change; QP-306 First-Off Approval; SOP-105 Product Changeover and Line Clearance

Records: FRM-PR-04 Changeover Record

6.0 PREPARATION

1. *Shift Lead:* Confirm the last pallet of the old size is complete and labelled.
2. *Operator:* Collect the change parts kit for the new size from the maintenance crib. Check every part against the list in the lid.
3. *Operator:* Clear the line of all bottles, caps, labels and cases of the old size, as described in SOP-105.
4. *Maintenance Technician:* Lock out FIL-2 and CAP-2 as described in SP-402.

7.0 CHANGE PARTS

1. *Maintenance Technician:* FIL-2: change the infeed screw, star wheels and centering bells. Set the nozzle height for the new bottle.
2. *Maintenance Technician:* CAP-2: change the bottle guides and adjust the head height.
3. *Operator:* CAP-2: both bottle sizes use the same 38 mm cap. Set the application torque to 15 in-lb, plus or minus 2. Check three bottles with the torque tester.
4. *Operator:* LAB-2: load the labels for the new size and set up as described in WI-212.
5. *Operator:* CPK-2: change the case magazine guides and select the pack pattern (6 per case for 2 L, 12 per case for 500 ml).
6. *Operator:* CNV-2: adjust the guide rails to the bottle width plus 3 mm on each side.
7. *Maintenance Technician:* Remove locks and tags as described in SP-402.

8.0 RESTART

1. *Operator:* Set up the filler for the new product as described in WI-210.
2. *QA Technician:* First-off approval as described in QP-306.
3. *Shift Lead:* Record start and end times and any missing parts on FRM-PR-04 Changeover Record.
4. *Operator:* Return the old change parts to the maintenance crib, clean and complete.

Note: Target changeover time: 45 minutes in either direction.

9.0 REVISION HISTORY

Rev	Date	Change
B	2021-03-15	Added target time.
A	2019-05-20	First issue.